

10 Networking Facts and Explanations

Fact 1: Network protocols define agreed rules for devices to format, send, and receive data. Shared rules let equipment and software from different makers communicate reliably.

Fact 2: An Internet Protocol (IP) address identifies a device interface on an IP network. Routers use destination IP addresses to move packets toward the correct network.

Fact 3: The Domain Name System (DNS) translates human-readable domain names into IP addresses. This lets people use names while network connections use numerical addresses.

Fact 4: A Media Access Control (MAC) address identifies a network interface on a local link. Ethernet switches use MAC addresses to forward frames within a local network.

Fact 5: A network switch connects devices on the same local network. It learns which MAC addresses are reachable through each port and forwards frames to the appropriate port.

Fact 6: A router connects separate IP networks and forwards packets between them. It checks routing information to choose a next hop for each destination.

Fact 7: Transmission Control Protocol (TCP) provides an ordered, reliable byte stream. It detects missing data and retransmits it, which suits applications that need complete delivery.

Fact 8: User Datagram Protocol (UDP) sends independent datagrams without guaranteeing delivery or order. Its lower overhead is useful when an application can tolerate loss or manage it itself.

Fact 9: Dynamic Host Configuration Protocol (DHCP) automatically provides network settings such as an IP address, subnet mask, gateway, and DNS server. This reduces manual device configuration.

Fact 10: Transport Layer Security (TLS) encrypts data in transit and helps verify a server identity. HTTPS uses TLS to protect communication between a web browser and a website.